

The Mathematics of Motion: Automated Discovery of Biomechanical Laws

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Abstract

There are presently no agreed upon models of human locomotion, but not for lack of trying. This not only limits the scientific understanding of human walking but also delays clinical innovation. Prior approaches range from linear regression to machine learning algorithms (e.g., random forests, support vector machines, deep learning, etc.), and first principle physical models of gait. However, each of those approaches face shortcomings. Some are overly simple; some lack physical, physiological, or biomechanical meaning; and some are too complex. Symbolic Regression (SR) offers a middle ground by automatically discovering interpretable, yet accurate, equations directly from data. To establish validity, we first re-discovered equations of well-known dynamical systems (the logistic map, the Lorenz system, and the passive dynamic walker). Then, we applied SR on 1,944 overground walking trials, deriving second-order differential equations for thigh, shank, and foot dynamics. Our models accurately predicted the lower limb segment angular accelerations (all $\bar{R}^2 \geq 0.95$). The discovered equations were derived from a single composite gait cycle (101 time points) yet generalized across thousands of gait trials. Crucially, these equations were transparent. Their terms can be mapped directly onto biomechanical interactions, allowing researchers to gain meaningful insights on gait coordination. Overall, our findings demonstrate that SR can identify interpretable equations that balance parsimony and accuracy. This balance affords SR as a transformative pathway for advancing motor control theory, guiding clinical rehabilitation, and shaping future biomechanical models.

Significance Statement

Modeling how human walk has challenged scientists for decades, with existing approaches either oversimplifying gait dynamics or producing black-box models that obscure meaning. Here, we demonstrate that symbolic regression – a method that discovers equations from data – provides a powerful middle ground. From the equivalent of a single gait cycle, we derived concise and generalizable equations that not only predicted gait with high accuracy but also can be mapped directly onto biomechanical components. This balance allows researchers to derive governing laws of locomotion, generate new hypotheses about motor control, and inform clinical and technological applications. By bridging simplicity and complexity, symbolic regression offers a transformative framework for understanding human movement and its broader applications in rehabilitations, assistive devices, and robotics.

Main Text

Introduction

Modeling and analyzing data are core tasks for all scientists. The situation is no different for researchers of human movement and physiology. Human movement is dynamic and takes on many forms that constitute activities of daily living, such as walking, talking, eating, bathing, and so on. Given the range and complexity of human movement, scientists have leveraged broad methodological approaches to describe, classify, predict, and control movement (1–4). Those methods range from traditional ordinary least squares regression and related techniques to time series analysis, cutting-edge deep learning, and the use of physical models that resemble human gait (1, 5–21). Each approach carries with it a unique set of benefits and limitations. In this manuscript, we showcase what we consider to be an under-used approach in human movement science known as *symbolic regression*. Briefly, symbolic regression uses various optimization algorithms to discover equations directly from data. Though originating in the 1990s (22), its adoption has been slow in all areas of science but has recently gained notoriety due to advances in computing and its ability to rediscover known equations of motion for complex dynamical systems as well as discover new ones (23–25). Below, we demonstrate this exciting technique in the

challenging domain of human gait to discover new equations of motion that govern the coordination and control of human movement (26).

The paper is organized as follows: First, we provide a summary of the various modeling approaches applied to human movements. Those typically fall into three categories, (1) statistical models such as ordinary least squares (OLS) regression, (2) ‘black box’ machine learning algorithms that often have a large number of internal parameters that are challenging for humans to interpret, and (3) simple physics-based models of bipedal locomotion that have been used as templates of gait (15, 27, 28). Then, we describe the symbolic regression paradigm. In this manuscript, we focus mainly on the genetic algorithm variant of symbolic regression but also mention other optimization schemes. Then, we conduct two data experiments, one simulated and one with laboratory gait data. In both cases, our goal is the same – to extract governing equations from data. Simulated data includes famous dynamical systems such as the logistic map and Lorenz system but also a simple physical model of bipedal gait, the passive dynamic walker model (15, 29–31). In the simulated data, the governing equations are known, while in the gait data, they are not. Lastly, we conclude with a discussion of the implications of our findings in terms of simulating data for model development, animation, robotics, interpretability, and, most importantly, developing new knowledge and hypotheses about human movement and bodily coordination.

Models of Movement

In a perfect world, all scientific models would be human interpretable. Interpretable, in this context, simply means that someone with sufficient domain knowledge could inspect the contents of a model, along with its predictions, and extract meaning or insight about the process the model is meant to represent. Unfortunately, that is not always the case. The rise of machine learning tools over the last decade or more has brought with it an immense number of powerful modeling approaches (32). However, machine learning-based models often lack interpretability, making it difficult to extract meaningful insights about the coordination underlying movement. On the one hand, simple models, such as random forests and support vector machines, do admit some interpretability through their relatively simple algorithms and measures of variable importance. Unfortunately, these models often (1) struggle to capture complex relationships in the data, (2) depend on the quantity and quality of the data, and (3) face issues with either over- or under-fitting (33, 34). On the other hand, neural network-based models, like long short-term memory models (LSTMs) and large language models (LLMs), are incredibly powerful for finding patterns in data. However, those models often require immense computational resources to appropriately capture the dynamics of the data (32). Another downside of those models is that they operate like ‘black boxes’ due to the sheer volume of parameters and their connections, failing to advance our understanding of the underlying rules that govern motor control (35). Ordinary least squares (OLS) regression, a more interpretable machine learning method, requires researchers to know all the relevant predictors and to decide the model structure in advance - all without knowing a system’s governing equations. Researchers may be tempted to add new variables or higher order polynomials to improve the fit, but such a process reduces the interpretability of the model and increases the chance of over-fitting. This leads to a trade-off between model complexity and interpretability, underscoring the demand for approaches that balance predictive power with the ability to extract insights and predictions about the principles of human movement.

Besides the previously mentioned data-driven, bottom-up approaches, there have been decades of work devoted to modeling human movement from a more top-down perspective. Quantifying the external work and mechanical forces involved in human walking provided foundational knowledge (12, 13). That knowledge introduced the inverted pendulum model of human gait and enhanced our understanding of gait mechanics (Figure 1A). The concept of ballistic walking (36), a perspective that treats phases of the gait cycle as ballistic motion, complemented the inverted pendulum model (Figure 1B) (18). Expanding on those ideas, passive dynamic walker models and their incrementally more realistic derivatives were featured prominently when developing theory and engineering efforts over the last several decades (18, 28, 37–40). For example, the *Simplest*

Walking Model...(15) is essentially two rigid legs connected at a hip joint, with point masses serving as the center of mass and feet (Figure 1C). When modeled on a negatively sloped floor, the model walks unaided and unpowered by nothing but the physics of the system under gravity. Furthermore, an increasingly negative floor slope modifies the dynamics of the walker through a series of bifurcations, ultimately leading to mathematical chaos, a commonly observed property of human gait (41). The appeal of such models is self-evident. They are derived from physical and mathematical principles and are often quite simple. As shown below, Garcia's (15) passive dynamic walker models can be boiled down to just two equations (see Simulation Experiments), of which the number and arrangement of terms that compose them relatively simple. While simplicity is key for interpretability, the simplicity of those models can also be a limitation because the models may overlook the mechanical and structural components required for biological locomotion. Top-down approaches often require strong assumptions and simplifications about the underlying mechanics, which leads to models that may not capture all aspects of real-world gait dynamics. Collectively, the limitations of the existing human movement models emphasize the demand for interpretable yet increasingly realistic models.

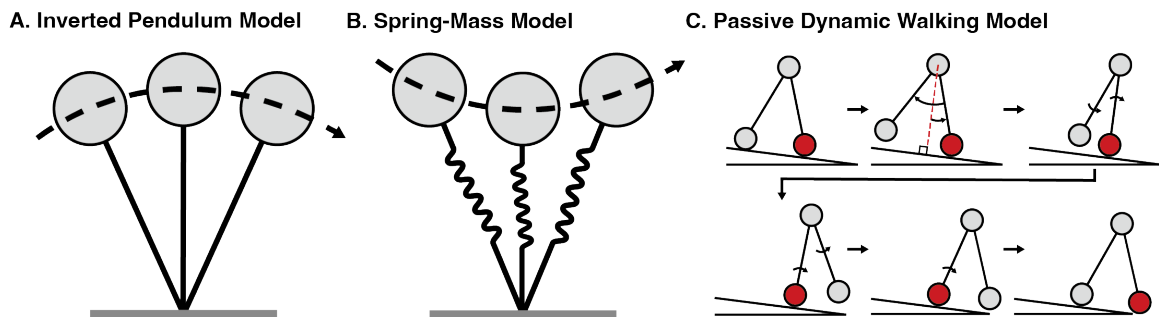


Figure 1. Models of Gait. (A) Inverted pendulum model is often used to describe walking gait. It consists of a massless rigid stance leg and a point mass representing the body's center of mass (COM) at the top. It captures how the exchange between the potential and kinetic energy propels the COM forward during stance. (B) Spring-mass model is frequently employed to describe running or jumping gait. It consists of a stance leg depicted as a mass-less spring, again with a point mass on the top of the spring. By including an elastic component, it captures the how energy is stored and released during stance, moderating the exchange between the potential and kinetic energy. (C) Passive dynamic walking model shows how bipedal walking can emerge from gravitational force when walking down a descending slope. It consists of two rigid legs that pivot on the body's COM with point masses at the feet. It captures how slope angle, which reflects the gravitational force, influences the bipedal gait dynamics. Depending on the slope angle, passive dynamic walker can exhibit stable, cyclic walking patterns or variable, chaotic walking patterns.

Symbolic Regression through Genetic Algorithms

As noted above, symbolic regression (SR) refers to a class of techniques that (re-) discover equations from data. The goal of such modeling is to discern equations that capture the patterns of the data while maintaining simplicity and interpretability. That is, the goal is to find a function, $y = f(X)$, where y is the outcome variable of interest, f is a mathematical function that relates a set of predictors, X , to the outcome variable y . The general steps needed to perform SR are as follows:

1. Decide on a set of input variables ($X = [x_1, x_2, \dots, x_p]$) that may predict some outcome variable of interest (y). Note that some variables, x_i , will often not be included in the model. At the outset, the contents of X are just a "catalogue" of potentially relevant variables and transformations. The variables will be those collected with laboratory instruments, and,

depending on the goal, could also include numerical derivatives of measured quantities. We are interested in discerning equations of motion, so that is the approach we take below.

2. Decide on a set of mathematical operators and functions that will serve as the 'genes' used in the genetic algorithm. These are arithmetic operators (e.g., +, -, $\times$, /) as well as basic mathematical functions (e.g., $\sin x$, $\cos x$, e^x). In practice, all mathematical functions are fair game. For example, one can also use floor, ceiling, mod, or even custom functions that combine operators, and so on. In the present work, we keep things simple (See Methods) by only using basic arithmetic operators, along with $\sin x$ and $\cos x$. Several constraints can also be set such as forbidding nested trigonometric functions, a prescription we followed as well.
3. Decide on a loss function by which to evaluate the performance of a given function (or a given expression) such as the root mean squared error or the mean absolute error. Note, though, that the analyst is free to use custom loss functions, even those that are not differentiable – differentiability is a typical requirement for loss functions in most machine learning contexts (e.g., the OLS solution and back propagation in neural networks).
4. Use a genetic algorithm to evolve mathematical expressions (encoded as trees, a bit like a flow diagram; Figure 2 right side) using principles of evolution such as crossover, competition, mutation, and migration. That is, one or more populations of expressions are initialized from the set of operators and functions determined in Step 2. Those expressions are evaluated based on their fitness with respect to the loss function determined in Step 3. The fitness level relative to the rest of the population determines which expressions 'die off' and which expressions 'live on' to pass their expressions to the next generation of equations. There are many approaches to making this determination. Those range from simple methods, like setting a threshold at the median fitness observed in the current generation, to complex ones involving elaborate tournament systems. Crossover occurs when two (or more) expressions exchange some proportion of their components. This exchange is often perturbed by a small fraction of mutations. In addition, when the algorithm involves multiple populations, a small proportion migrates and competes with other expression populations.
5. Repeat the evolutionary process in Step 4 many times until a satisfactory fit of the data is achieved. Sometimes this convergence is quite fast (on the order of seconds). Sometimes the process is slow (on the order of a week or more). What constitutes "satisfactory" is easy to define when equations are known – the SR either returns the correct equation or it does not. A satisfactory fit is less concrete in the context of laboratory data but is typically decided based on *post hoc* analysis of the fitness of returned equations as a function of the equation's complexity. We show one such method below.

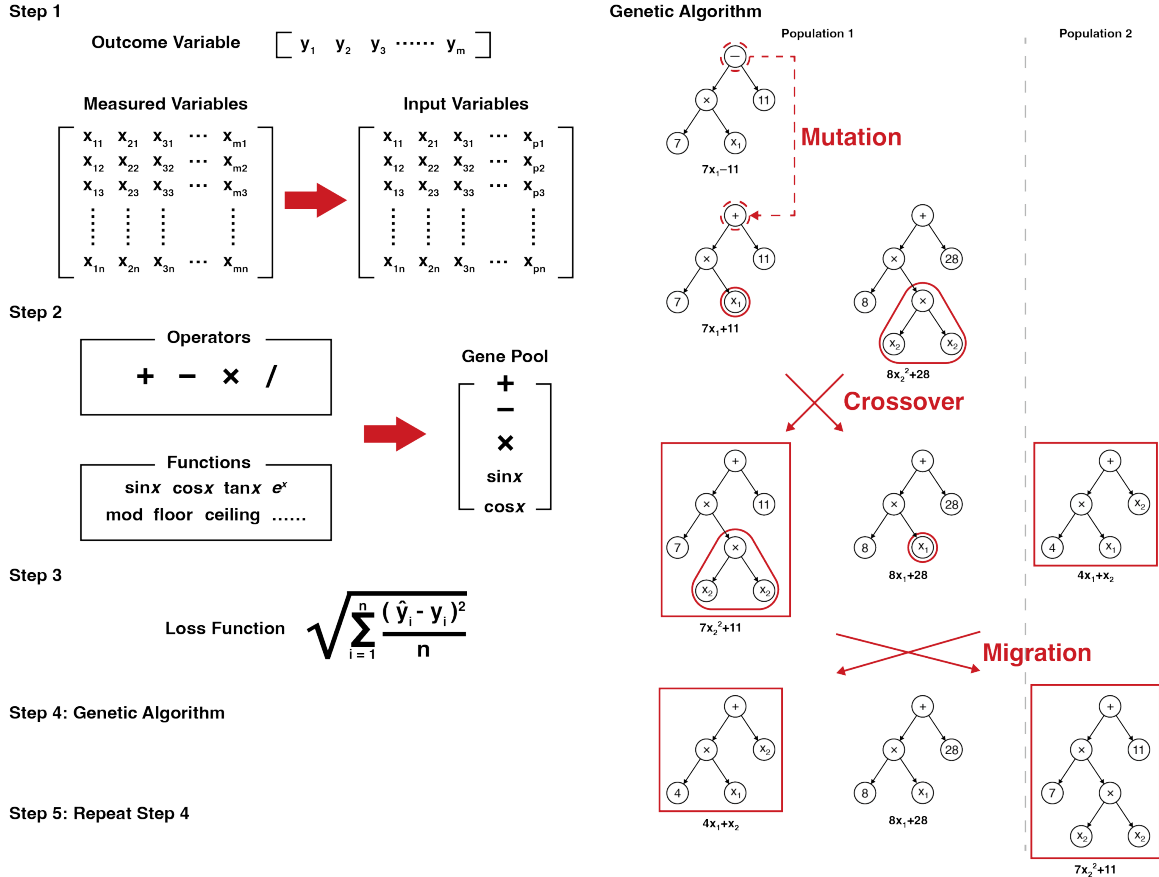


Figure 2. Five steps of symbolic regression. (Right) 5 general steps of symbolic regression. Step 1: Decide on a set of input variables and outcome variable. Step 2: Decide on a set of mathematical operators and functions that will serve as the ‘genes’ used in the genetic algorithm. Step 3: Decide on a loss function to evaluate the performance of a given function (or expression) . Step 4: Use a genetic algorithm to evolve mathematical expressions using principles of evolution such as crossover, competition, mutation, and migration. Step 5: Repeat the evolutionary process in Step 4 until a satisfactory fit of the data is achieved. (Left) Graphical representation of genetic algorithm.

Although relatively simple, those five steps are remarkably powerful in extracting equations from data. For example, Schmidt and Lipson (23) were able to recover equations of motion from a number of physical systems with relevance to movement, such as single and double pendula as well as harmonic oscillators. In the experiments described below, we demonstrate the potential for SR to model human gait with relatively simple equations. Remarkably, we also show that those simple equations preserve characteristic dynamics that emerge over many gait cycle repetitions.

Before applying SR to extract equations of motion from human gait data, we first validated its ability to rediscover equations using simulated data. We synthesized data from known dynamical systems, including the logistic map (42), the Lorenz system (30), and a passive dynamic walker model (15). We then applied SR to those data, successfully rediscovering the original system equations and accurately predicting the simulated data.

Following the validation, we applied SR to human gait data from the NONAN Gaitprint dataset (43) to unveil the equations of motion governing human walking (Figure 3). We computed the typical gait kinematics of a single gait cycle, employing SR to model the equations of motion. Right thigh, shank, and foot angular accelerations were the outcome variables, with bilateral angles and angular

velocities of the thigh, shank, and foot serving as predictors. We first generalized the discovered equations of motion to evaluate their accuracy in predicting individual gait kinematics. Then, we estimated the coefficients and the intercepts that best fit each equation to each walking trial using OLS regression. We predicted bilateral thigh, shank, and foot angular acceleration time series, based on the estimated coefficients and intercepts. Finally, we evaluated the accuracy of the predicted time series and the ability of the equations to reproduce the patterns of variability seen across gait cycles.

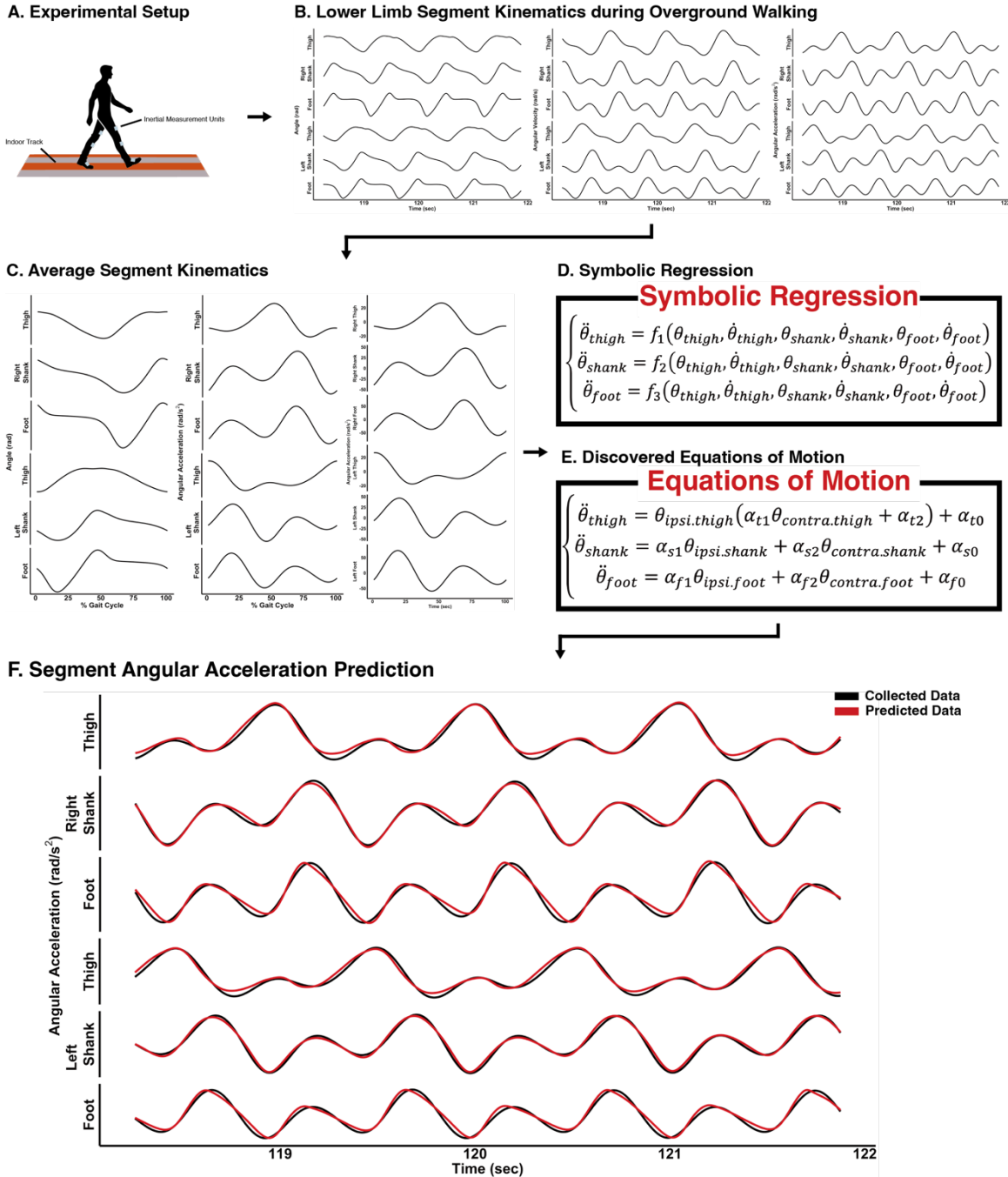


Figure 3. Overview of Empirical Experiment. (A) Experimental setup. Each subject were equipped with inertial measurement units to collect gait kinematics. (B) Lower limb segment kinematics during

overground walking. Thigh, shank, and foot segment angle, angular velocity and acceleration were taken for following procedures. (C) Average segment kinematics. Segment kinematics were ensemble averaged across all subjects and all trials (a total of 1944 trials). (D) Symbolic regression. Average right thigh, shank, and foot segment angular accelerations were set as the outcome variable, whereas bilateral thigh, shank, and foot segment angles and angular velocities were set as the input variables. (E) Discovered equations of motion. From a list of discovered equations of motion, we selected an equation of motion for each segment based on our selection criteria. (F) Segment angular acceleration prediction. Segment angular accelerations were predicted using ordinary least squares regression. Black lines indicate a sample of collected data. Red lines indicate a sample of predicted data.

Results

Simulation Experiments

To validate the capability of SR, SR was performed on simulated dynamical systems, including the logistic map, the Lorenz system, and the passive dynamic walker model. From the list of equations given by SR, f_{SR} , those balancing parsimony and accuracy were selected (see supplementary materials for selection criteria). The following f_{SR} s were discovered most frequently for each system (see Table S1 for other equations).

Logistic Map

$$x_{t+1} = 3.6x_t(1 - x_t)$$

Lorenz System

$$\begin{cases} dx/dt = 10(y - x) \\ dy/dt = x(28 - z) - y \\ dz/dt = xy - \frac{8}{3}z \end{cases}$$

Passive Dynamic Walker Model

$$\begin{cases} \ddot{\theta} = \sin(\theta - 0.014) \\ \ddot{\phi} = \ddot{\theta} + \dot{\theta}^2 \sin(\phi) - \cos(\theta - 0.014) \sin(\phi) \end{cases}$$

For the logistic map and the Lorenz system, in most cases (logistic map: 100%; Lorenz system dx/dt : 100%; dy/dt : 100%; dz/dt : 90%), SR rediscovered the exact equations. For the passive dynamic walker model, SR successfully rediscovered the $\ddot{\theta}$ equation ($\ddot{\theta}$: 100%) but struggled with the $\ddot{\phi}$ equation ($\ddot{\phi}$: 15%).

f_{SR} accuracies were evaluated by numerical integration and comparison with original time series (see supplementary materials, Figure S1). For the logistic map and the passive dynamic walker model, f_{SR} s flawlessly replicated the original time series ($R^2 > 0.999$). Whereas, for the Lorenz system, f_{SR} s led to a less satisfying outcome despite f_{SR} s being almost identical to the original Lorenz system (R^2 of $x = 0.030 \sim 0.146$; R^2 of $y = 0.030 \sim 0.142$; R^2 of $z = 0.103 \sim 0.213$). The low R^2 obtained at all axes is due to the Lorenz system's sensitivity to initial conditions, which is elaborated in the discussion section.

Empirical Experiments

To discover the equations of motion describing human gait, SR was implemented on thigh, shank, and foot angular acceleration time series from 1944 overground walking trials. From the list of equations given by SR, f_{SR} s were selected (see supplementary materials), and then generalized

by substituting the coefficients and constants with algebraic terms (see supplementary materials for details). The most frequently discovered f_{SR} s were:

Thigh

$$\ddot{\theta}_{thigh} = \theta_{ipsi.thigh}(\alpha_{t1}\theta_{contra.thigh} + \alpha_{t2}) + \alpha_{t0}$$

Shank

$$\ddot{\theta}_{shank} = \alpha_{s1}\theta_{ipsi.shank} + \alpha_{s2}\theta_{contra.shank} + \alpha_{s0}$$

Foot

$$\ddot{\theta}_{foot} = \alpha_{f1}\theta_{ipsi.foot} + \alpha_{f2}\theta_{contra.foot} + \alpha_{f0}$$

θ and $\ddot{\theta}$ represents segment angles and angular accelerations, respectively. 'Ipsi' and 'contra' refer to ipsilateral and contralateral inputs relative to the outcome variable (e.g., for the right thigh angular acceleration, $\theta_{ipsi.thigh}$ and $\theta_{contra.thigh}$ would refer to right thigh segment and left thigh segment angle, respectively). Coefficients α_{t1} , α_{t2} , α_{s1} , α_{s2} , α_{f1} , α_{f2} moderate the influence of input variable kinematics on the outcome variable kinematics. α_{t0} , α_{s0} , α_{f0} are the intercepts of each outcome variable kinematics. Alphabetical subscripts, t , s , and f , denote thigh, shank, and foot, respectively. Numerical subscripts reflect order of appearance in the equations, except for 0s which were assigned to the intercepts. In most SR trials, within each segment, identical equations of motion were selected as f_{SR} (see the number of instances per f_{SR} in supplementary materials, Table S2).

To assess f_{SR} accuracies, angular accelerations were predicted for each trial by fitting the f_{SR} with OLS regression. The predicted and the original angular acceleration time series were then compared via R^2 (Figure 4). The discovered f_{SR} captured the original time series with high accuracy ($R^2_{right thigh}$: 0.958 ± 0.044 ; $R^2_{left thigh}$: 0.957 ± 0.045 ; $R^2_{right shank}$: 0.981 ± 0.013 ; $R^2_{left shank}$: 0.980 ± 0.013 ; $R^2_{right foot}$: 0.969 ± 0.017 ; $R^2_{left foot}$: 0.969 ± 0.017). For the sake of readability, the in-depth mathematical interpretations of each equation have been relegated to the supplementary materials.

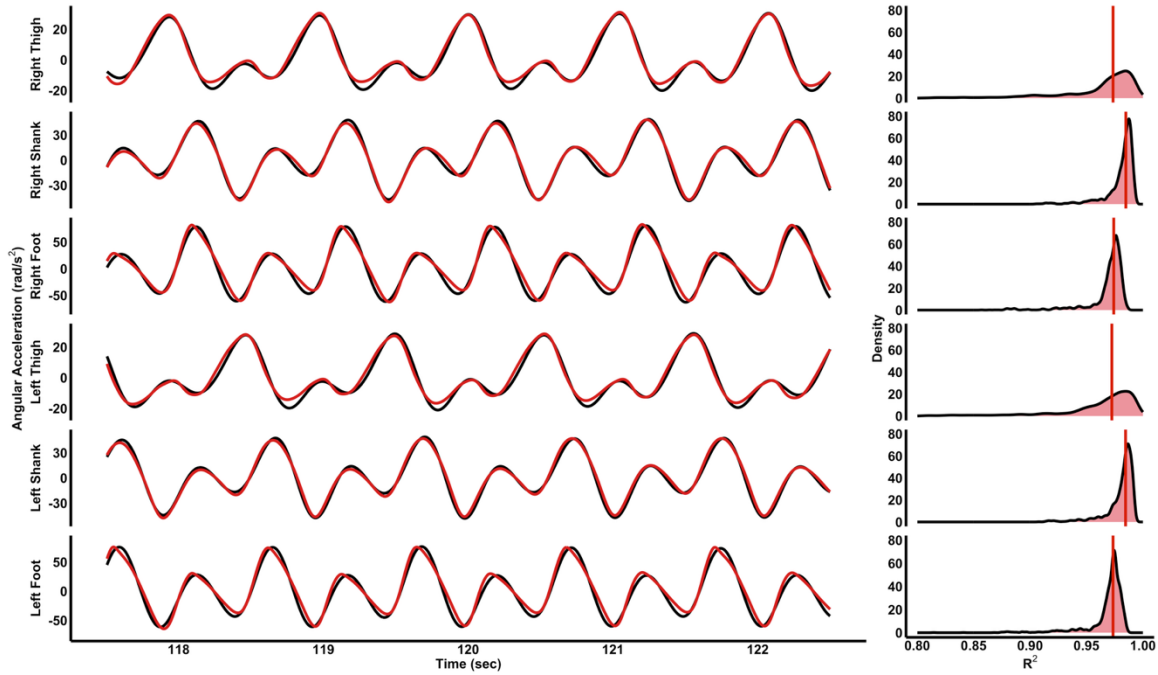


Figure 4. Segment Angular Acceleration Prediction. (Left) Example segment angular acceleration time series and its prediction based on the equations of motion discovered by symbolic regression for each segment. Black lines indicate the example segment angular acceleration time series. Red lines indicate the predicted segment angular acceleration time series. (Right) Probability density function of prediction accuracy for each segment. Red vertical lines indicate the median prediction accuracy.

Along with predicting time series via OLS regression, we investigated whether the predicted angular acceleration time series preserved the rich pattern of gait variability. Hurst exponents ($H_{original}$ and $H_{predicted}$; the details on Hurst exponent can be found in the methods section and supplementary materials) were quantified from the original and predicted time series. Bland-Altman analyses were then conducted to examine the agreement between $H_{original}$ and $H_{predicted}$ (see supplementary materials, Figure S2). The Bland-Altman analyses revealed no systematic differences between $H_{original}$ and $H_{predicted}$ ($\Delta H_{right\ thigh} = 0.040 \pm 0.037$; $\Delta H_{left\ thigh} = 0.037 \pm 0.041$; $\Delta H_{right\ shank} = -0.040 \pm 0.064$; $\Delta H_{left\ shank} = -0.051 \pm 0.082$; $\Delta H_{right\ foot} = -0.062 \pm 0.045$; $\Delta H_{left\ foot} = -0.065 \pm 0.052$). Put succinctly, the equations of motion, discovered via SR, reliably captured the variability underlying human gait kinematics.

Discussion

In this study, we derived the equations of motion for human walking by implementing SR with lower limb kinematic data. Prior to employing SR on human gait kinematics, we showed that SR can correctly rediscover the governing equations of dynamical systems, including the passive dynamic walker model. By performing SR on empirical gait data, we obtained equations of motion that consistently had low equation complexity for all lower limb segments. Based on a follow-up OLS regression, all equations of motion accurately predicted both the original segment angular acceleration time series and the cycle-to-cycle dynamics. Below, we interpret the discovered equations of motion and discuss features and novel implications of SR.

A Biomechanical Interpretation of Learned Gait Equations of Motions

In the introduction, we defined ‘interpretable models’ as models that 1) allow individuals with sufficient knowledge to study the model’s contents and predictions, and 2) provide insight into the

processes the model represents. In the supplementary materials, we derived the mathematical meaning of each coefficient and constant in each f_{SR} . However, an integrative explanation within the context of human walking is necessary to fully appreciate these equations.

The discovered equations are structurally simple, comprising only the sums and products of two segment angles and a constant. As demonstrated in supplementary material, interpreting the sums and products is relatively straightforward. The sum of two segment angles represents the linear relationship between each segment angle and segment angular acceleration. The product of two segment angles indicates the dynamics of one segment moderating the linear relationship between the other segment angle and segment angular acceleration. Based on that approach, we interpret model coefficients in terms of gait kinematics to clarify the model's representation of overground walking.

Thigh

$$\begin{cases} \ddot{\theta}_{ipsi.thigh} = \theta_{ipsi.thigh} \cdot (\alpha_{t1} \cdot \theta_{contra.thigh} + \alpha_{t2}) + \alpha_{t0} \\ \ddot{\theta}_{contra.thigh} = \theta_{contra.thigh} \cdot (\alpha_{t1} \cdot \theta_{ipsi.thigh} + \alpha_{t2}) + \alpha_{t0} \end{cases}$$

The constant α_{t0} equals $\ddot{\theta}_{ipsi.thigh}$ at mid-stance and mid-swing, when the ipsilateral thigh is perpendicular to the walking surface ($\theta_{ipsi.thigh} = 0$). α_{t2} encodes the relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{ipsi.thigh}$ when the contralateral thigh is perpendicular to the surface ($\theta_{contra.thigh} = 0$; i.e. during the mid-stance and mid-swing phase of the contralateral leg). Lastly, α_{t1} scales the influence of $\theta_{contra.thigh}$ on the relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{ipsi.thigh}$, playing more significant role as the contralateral thigh moves away from vertical position. Conversely, as the contralateral thigh becomes perpendicular to the surface, the impact of α_{t1} diminishes, leaving the relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{ipsi.thigh}$ dependent on α_{t2} . Furthermore, based on the mean values of α_{t1} and α_{t2} ($\alpha_{t1} = -70.437$, $\alpha_{t2} = -36.383$) and the range of $\theta_{contra.thigh}$ ($[-0.36, 0.51]$), we can infer that $\alpha_{t1} \cdot \theta_{contra.thigh} + \alpha_{t2}$ is consistently negative, characterizing the temporal variation of the anti-phasic relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{ipsi.thigh}$.

Shank

$$\begin{cases} \ddot{\theta}_{ipsi.shank} = \alpha_{s1}\theta_{ipsi.shank} + \alpha_{s2}\theta_{contra.shank} + \alpha_{s0} \\ \ddot{\theta}_{contra.shank} = \alpha_{s1}\theta_{contra.shank} + \alpha_{s2}\theta_{ipsi.shank} + \alpha_{s0} \end{cases}$$

In contrast to the thigh, α_{s0} represents $\ddot{\theta}_{ipsi.shank}$ when the weighted sum of the bilateral shank angles ($\alpha_{s1}\theta_{ipsi.shank} + \alpha_{s2}\theta_{contra.shank}$) equals 0. The net cancellation of the bilateral shank angles suggests that α_{s0} reflects the passive dynamics of shank induced by external forces, such as ground reaction forces delivered through the ankle joint, and the hip-generated torque transmitted through the knee joint. Given the mean values of coefficients, α_{s1} and α_{s2} ($\alpha_{s1} = -83.076$, $\alpha_{s2} = -46.620$), these coefficients delineate the independent anti-phasic relationship between $\theta_{ipsi.shank}$ and $\ddot{\theta}_{ipsi.shank}$, and $\theta_{contra.shank}$ and $\ddot{\theta}_{ipsi.shank}$. Specifically, α_{s1} (or α_{s2}) portrays the negative linear relationship between $\theta_{ipsi.shank}$ (or $\theta_{contra.shank}$) and $\ddot{\theta}_{ipsi.shank}$ when the contralateral shank (or ipsilateral shank) is perpendicular to the surface. This occurs shortly after the ipsilateral (or contralateral) leg's toe off. This further suggests that α_{s1} (or α_{s2}) quantifies the amplitude of the

anti-phasic relationship between $\theta_{ipsi.shank}$ (or $\theta_{contra.shank}$) and $\ddot{\theta}_{ipsi.shank}$ driven by the propulsion of the ipsilateral (or contralateral) leg.

Foot

$$\begin{cases} \ddot{\theta}_{ipsi.foot} = \alpha_{f1}\theta_{ipsi.foot} + \alpha_{f2}\theta_{contra.foot} + \alpha_{f0} \\ \ddot{\theta}_{contra.foot} = \alpha_{f1}\theta_{contra.foot} + \alpha_{f2}\theta_{ipsi.foot} + \alpha_{f0} \end{cases}$$

In reference to the foot, $\ddot{\theta}_{shank}$ and $\ddot{\theta}_{foot}$ trajectories are akin to each other and their f_{SR} share identical forms. Thus, we do not repeat the detailed explanations for f_{SR} of $\ddot{\theta}_{foot}$. Instead, we briefly summarize the implications of each coefficient and constant within f_{SR} of $\ddot{\theta}_{foot}$ in the context of human walking. Like α_{s0} , α_{f0} may display relatively passive foot dynamics caused by the external forces during the late-swing phases (e.g. hip flexion torque transferred through knee and ankle joints). The coefficients, α_{f1} and α_{f2} each would reflect the anti-phasic relationship between $\theta_{ipsi.foot}$ and $\ddot{\theta}_{ipsi.foot}$, and $\theta_{contra.foot}$ and $\ddot{\theta}_{ipsi.foot}$ triggered by the propulsion of their respective leg.

Gaining Insights from Discovered Equations of Motion

The interpretable equations derived through symbolic regression accurately predicted gait kinematics with median R^2 exceeding 0.95. Remarkably, this was achieved with only a single gait cycle's worth of data. Importantly, our approach preserved the temporal structure of gait variability in the original segment angular acceleration time series (Figure S2). Conventionally, quantifying patterns of gait variability has been challenging due to “data-hungry” algorithms (44, 45). Once further refined, our method should enable researchers to predict superficial patterns in gait (i.e., gait kinematics and kinetics) along with deeper aspects of gait dynamics (i.e., structure of gait variability) from just a few strides, bypassing conventional limitations.

Beyond prediction, the true strength of interpretable mathematical models emerges when they yield theoretical insights into the underlying dynamics. Understanding the dynamics entails exploring how behavior changes in response to various constraints. Constraint-led behaviors can be understood by 1) tuning the predictors and coefficients in a model to make theoretical predictions, and 2) conducting experiments driven by those predictions to validate how the system's behavior responds to specific constraints. For instance, in passive dynamic walker models, hypotheses about mathematical chaos in bipedal gait were made by tuning the slope (γ) and other coefficients. A variant of the passive dynamic walking model(38) revealed that increasing the hip joint actuation coefficient transitioned a chaotic walking pattern—which should happen when the negative slope increases—into a strictly periodic walking pattern. Translating this to human gait, the hip joint actuation would be equivalent to activation of hip flexors and extensors. Hence, one can experimentally control the activation of the hip flexors and extensors and observe how the gait patterns are modified to test the prediction made through models. Similarly, the equations of motion discovered in this study allow predictions about gait dynamics by tuning the model coefficients and constants. For example, increasing or decreasing α_{t0} , the ipsilateral thigh angular acceleration at mid-stance and mid-swing, will alter the gait dynamics, allowing researchers to establish hypotheses on how gait dynamics change as a function of the constraint that is relevant to α_{t0} . Furthermore, if we claim that “ α_{t0} is associated with the hip joint torque generation during walking”, we can experimentally control the hip joint torque generation during walking and test our hypotheses. Indeed, one can iteratively apply symbolic regression to newly obtained empirical data, compare the newly discovered equations of motion, and find new relevant predictors and interactions among predictors. This iterative cycle—deriving equations, testing hypotheses experimentally, and refining equations based on empirical insights—will progressively develop more realistic and integrative models. Ultimately, the interplay between interpretable mathematical

models and rigorous, hypothesis-driven experiments will promote a mutually beneficial cycle, enhancing both theoretical understanding and practical applications in human movement science.

The Impact of Computational Sensitivity on Prediction

The f_{SR} for the logistic map was mathematically equivalent to the original logistic map, precisely reproducing the original logistic map time series. However, that was not the case for the Lorenz system. Although SR correctly identified all three equations of the Lorenz system, numerical integration of f_{SR} produced trajectories that diverged from the original (see supplementary materials, Figure S1). Mathematically chaotic dynamical systems, such as the Lorenz system, exhibit sensitivity to initial conditions, meaning even miniscule differences in the initial state or the coefficients (e.g. α in the logistic map and β , σ , ρ in the Lorenz system) can result in exponentially diverging trajectories (46). Hence, marginal differences between coefficients (control parameters) likely caused the major disparity between the original and the predicted Lorenz system time series. How can one resolve the drop in prediction accuracy due to the sensitivity to initial conditions?

The issue can be attenuated by substituting the coefficients and constants in the equations with more meaningful values based on predefined simplifying assumptions. Some examples of simplifying assumptions include rounding the coefficients and the constants to their closest integers (e.g., 0.999 to 1) and substituting the coefficients and the constants with the closest, yet meaningful, rational (e.g., 1.333 to $4/3$) and/or irrational numbers (3.142 to π). If one follows such simplifying assumptions, the rediscovered equations of the Lorenz system can be simplified to match the original equations (rounding 9.996 to its closest integer, 10, and substituting 2.666 with the closest rational number, $8/3$), mitigating the divergence and improving the accuracy of the predictions.

Of course, there can be exceptions in which simplifying assumptions may not fulfill their purpose. When ‘true’ coefficients and constants are not simple values (e.g., integers, meaningful rational/irrational numbers), simplifying assumptions might not be sufficient and in fact harm the accuracy of the discovered equations. In such cases, researchers cautiously decide on their set of simplifying assumptions or pursue different approaches to finding the ‘true’ values of the coefficients and constants. For example, one can include specific values that are relevant to the nature of the data as predictors (e.g., if the data is periodic, π ; if the data is related to gravitational force, gravitational acceleration, g ; or even Euler’s number, e , if the data is related to the secrets of the universe). Additionally, one can employ diverse optimization techniques to identify the most accurate values for the coefficients and the constants. Although some symbolic regression algorithms already include optimization procedures, adding optimization tools that are suitable for the characteristics of the given data may yield coefficients and constants different from those found through symbolic regression. Overall, researchers should be mindful about the coefficients and constants, judging whether the values can be improved, and thoughtful about the approach they take in attenuating any potential impact of computational sensitivity.

Variability within Symbolic Regression: Error? Or Information?

Due to the probabilistic nature of SR, the equations discovered at a given level of complexity may differ across trials. In other words, the final solutions obtained for each SR trial may vary, potentially resulting in varying interpretations and predictions of a given phenomenon. Thus, conclusions shouldn’t be drawn based on only a single trial of SR. Instead, SR should be conducted in multiple bouts, and the biologically and/or physically soundest conclusion should be chosen from amongst the candidate solutions.

The variation across different SR trials also suggests other merits. Just as gait variability is not merely random fluctuation between gait cycles, the variability in SR outcomes may not simply stem from the probabilistic nature of SR’s algorithm. Instead, it could offer additional insights into the representation of the system’s dynamics. Among the dynamical systems employed in this study, the equations with simpler forms (logistic map, Lorenz system, passive dynamic walking model $\hat{\theta}$) had minimal variability across SR trials. Conversely, the equation with a more complex form

(passive dynamic walking model $\dot{\phi}$) showed higher variability across SR trials. This difficulty could, of course, mean that it was more difficult to discover the $\dot{\phi}$ equation but could also imply that there is more than one way to explain the swing leg kinematics of passive dynamic walker ($\dot{\phi}$ refers to the swing leg segment angle acceleration). A similar principle applies to language—simple ideas can be expressed in only a few ways, while more complex ideas allow for a greater variety of descriptions (47). Likewise, the greater variability in SR outcomes may reflect the presence of multiple valid representations of the system's dynamics, rather than algorithmic inadequacy to readily discover a single correct equation.

Regarding the equations of motion during gait, the variability of the equations across 20 SR trials was minimal, with only one of the equations containing an additional component. This may be because averaging gait kinematics across all subjects and gait cycles removed most of the individual gait characteristics, potentially masking subject-specific gait characteristics. We hypothesize that SR outcomes may be less variable when the dynamics are simpler but more variable when the dynamics are complex. To test this idea, the SR outcome variability should be evaluated for well-known dynamical systems with different levels of complexity in their dynamics and governing equations. Experimentally, SR can be repeatedly performed on data from groups with distinct complexities, as quantified by nonlinear measures such as Hurst exponent. For example, younger adults and older adults are known to exhibit differing gait complexities, with aging often associated with loss of gait complexity (48, 49). If our hypothesis holds, comparing SR outcome variability across age groups could provide insights into age-related changes in the context of motor control. If that is the case, then attention should be paid not only to the discovered equations per se but also to the variability across those equations, as this may provide insight into the system's degeneracy and ability to explore multiple movement solutions.

Implications of Symbolic Regression in Human Movement

SR bridges traditional statistical approaches, such as OLS regression, and modern modeling techniques like deep learning by providing interpretable, accurate, and efficient human movement models. Its utility spans theoretical exploration and diverse practical applications. By identifying the key predictors and the interactions among them without a pre-specified structure, SR allows researchers to gain novel insights on how movements are coordinated under various conditions. For instance, researchers can use SR to model how environmental constraints like uneven terrain or physiological constraints like muscle fatigue alter movement. Such models will facilitate hypothesis-driven experiments, enhancing our understanding of human motor control and learning.

The parsimonious nature of SR-based models makes them appealing in computationally intensive domains, including robotics and human-machine interactions. In humanoid robotics, SR-derived models can simulate human-like movement with reduced computational load, enabling shorter response times and more realistic motions. Furthermore, SR will improve the design of exoskeletons and prosthetics by offering more intuitive and biomechanically informed support. Clinically, SR-based models can significantly advance diagnostics and rehabilitation strategies by precisely capturing movement patterns and their variability. For instance, modeling individuals' gait variability will help clinicians detect early markers of movement disorders and design interventions that are tailored to the specific biomechanical needs identified by the models. By bridging theoretical insights, engineering solutions, and clinical practices, SR will offer a transformative approach to understanding and modeling human movement. Its potential to balance interpretability and accuracy establishes it as a cornerstone for future advancements across disciplines.

Limitations and Future Directions

Despite its promise for modeling human movement, SR has some limitations. First, its probabilistic nature introduces variability in the discovered models, complicating the selection of the “best” model and necessitating cautious evaluation across multiple trials. Additionally, SR's success heavily depends on the quantity and quality of data. Insufficient or noisy data can lead to overfitting or overly simplistic models, limiting their real-world applicability. Meticulous data curation and

preprocessing are thus critical for ensuring robust and reliable SR outcomes. Lastly, our current SR framework does not integrate domain-specific constraints such as biomechanical principles or physical laws, potentially resulting in models that are mathematically valid but physiologically implausible.

To fully harness SR's potential in human movement science, future research must address these limitations and explore new opportunities. One promising avenue is developing our SR approach tailored to biomechanics, which involve embedding physical laws such as conservation of momentum and physiological constraints like joint range of motion into the framework. Integrating these principles would ensure biologically plausible and realistic models. Additionally, hybrid approaches that combine SR with machine learning, such as neural networks, could leverage the strengths of both methods—SR's interpretability and machine learning's predictive accuracy. Neural networks could be used to extract structural insight from the data, which is then fed into SR to obtain interpretable models. This approach allows handling noisy data and discovering complex mathematical models while maintaining interpretability (25, 50). Expanding the application of SR to diverse movements, such as lifting, running, or jumping, and to populations with varying characteristics, such as older adults or individuals with movement disorders, would provide comprehensive understanding of movement dynamics and variability across contexts. Finally, validating SR-derived models in clinical and real-world settings will confirm their practical relevance. For example, applying SR to clinical movement datasets could offer valuable insights into the progression of neurological disorders or the efficacy of surgical and therapeutic interventions. Addressing the limitations and continuing to explore innovative approaches will solidify SR as a versatile tool for human movement modeling.

Conclusion

In this study, we modeled human gait using simple equations discovered using symbolic regression. Prior to modeling human gait, we posited that symbolic regression would be capable of rediscovering the governing equations of two dynamical systems and the passive dynamic walker. Then, we implemented symbolic regression on overground walking data collected on young adults to discover potential equations of motion for the lower limb segments. Using an ordinary least squares regression approach, the discovered equations accurately predicted the original gait kinematics and reliably predicted the temporal structure of gait variability in the original gait kinematics. The equations of motion discovered through symbolic regression were not only accurate but also remarkably simple and biologically interpretable, making it an appealing tool for modeling human gait. This study established symbolic regression as an accessible yet powerful tool for unveiling the principles of human movement dynamics. From understanding motor control to innovating rehabilitation and human-machine interaction, symbolic regression will play a vital role in opening new avenues for studying and applying the principles of human movement.

Materials and Methods

In this section, we only introduce the general workflow. Our study consisted of two experiments, 1) a simulation experiment to validate SR's ability to rediscover known equations of well-studied dynamical systems, 2) an empirical experiment to discover equations of motion of human walking by applying SR to a large public gait dataset. The following sections explain the general flow of each experiment and how the performance of SR on each dataset was examined. Detailed descriptions of the methods (e.g., equation selection criteria, simplifying assumptions) can be found in the supplementary materials.

Simulation Experiment

Prior to implementing SR on human movement data, we first tested its ability to rediscover the equations of well-known dynamical systems, the logistic map and Lorenz system, and a physical

model of bipedal gait, the passive dynamic walker model (PDW). The equations used to simulate each system are shown below (see details in supplementary materials, Table S3).

Logistic Map

$$x_{t+1} = 3.6x_t(1 - x_t)$$

Lorenz System

$$\begin{cases} dx/dt = 10(y - x) \\ dy/dt = x(28 - z) - y \\ dz/dt = xy - \frac{8}{3}z \end{cases}$$

Passive Dynamic Walker Model

$$\begin{cases} \ddot{\theta} = \sin(\theta - 0.014) \\ \ddot{\phi} = \ddot{\theta} + \dot{\theta}^2 \sin(\phi) - \cos(\theta - 0.014) \sin(\phi) \end{cases}$$

SR was performed on the first 1000 data points for the logistic map and Lorenz system, while for the PDW, SR was applied on the average gait kinematics. The average gait kinematics of PDW was computed using an ensemble average across every gait cycle (the details on the SR parameters used in this experiment can be found in Table S4 and S5). SR returns a list of equations that predicts the data most accurately at each equation's complexity (or size). From the equation list, a single equation, f_{SR} , was selected according to predefined criteria (see supplementary materials for selection criteria). 20 SRs were conducted on each component of every dynamical system and the PDW.

f_{SR} s were simplified based on some simplifying assumptions (the details on the simplifying assumptions can be found in supplementary materials). Then, f_{SR} s were compared with the original models to verify whether they possess equivalent terms, coefficients, and constants. In addition, to quantitatively assess the prediction accuracies of f_{SR} s, f_{SR} s were numerically integrated. Finally, R^2 between the originally simulated data and the numerically integrated data were computed.

Empirical Experiment

In the following experiment, we employed the methods investigated above in the context of human movement data. We extracted 112 healthy participants (young adult: 35; middle adult: 36; older adult: 41; female: 67; detailed subject characteristics can be found in supplementary materials Table S6) from the NONAN Gaitprint database, a public online repository of overground walking data (43, 51). Each participant was equipped with sixteen Noraxon Ultium Inertial Measurement Units (IMUs; Noraxon, Inc., Scottsdale, AZ) that recorded gait kinematics at a rate of 200Hz. Each participant was tasked with completing nine four-minute overground walking trials at a comfortable pace they could maintain for thirty minutes (See (43, 51) for additional details). In total, 1944 overground walking trials were used for SR.

From each trial, we extracted bilateral thigh, shank, and foot segment angle time series in the sagittal plane. We differentiated the segment angle time series to obtain the segment angular velocity and acceleration time series. Then, segment angle, angular velocity, and angular acceleration time series were segmented into each gait cycle and ensemble averaged across all walking trials and gait cycles to retrieve the typical gait kinematics of a single gait cycle. More information on how the gait kinematics data was processed can be found in the supplementary materials.

To derive the equations of motion from the typical gait kinematics, SR was applied using the right thigh, shank, and foot angular accelerations as outcome variables. All segment angles and angular velocities were used as the predictors (the details on the SR parameters used in this experiment can be found in supplementary materials, Table S7). A single equation, f_{SR} , was selected from the

list of equations returned from SR using the same criteria as the simulation experiment (the details on the equation selection criteria can be found in supplementary materials). 20 SRs were conducted on each segment's angular acceleration.

f_{SR} s were generalized through a standard process. Ordinary least squares (OLS) regression was then applied to estimate the coefficients and the intercepts that best fit each equation to each walking trial. Based on the estimated coefficients and intercepts, bilateral thigh, shank, and foot angular acceleration time series were predicted ($\hat{\theta}$). Goodness of fit was evaluated by calculating R^2 between the original segment angular acceleration time series ($\ddot{\theta}$) and $\hat{\theta}$. Supplementary materials provide an example of the process.

Furthermore, we investigated whether $\hat{\theta}$ preserved the variability of $\ddot{\theta}$ across gait cycles. Gait variability across gait cycles was quantified using the Hurst exponent (H), which measures the persistence of time series and is known to be useful in capturing the rich patterns within gait variability (52, 53). By identifying the peak angular accelerations in every gait cycle and subsequently computing the time durations between those peaks, a sequence of inter-peak intervals was generated for H estimation. The Hurst-Kolmogorov process, a Bayesian technique-based estimation of H , was used. See Koutsoyiannis, 2003, Likens et al., 2023; Tyralis & Koutsoyiannis, 2014 for additional details. Finally, Bland-Altman analyses were conducted to examine the agreement between H of the original ($H_{original}$) and the predicted ($H_{predicted}$) inter-peak interval time series.

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Supporting Information for

The Mathematics of Motion: Automated Discovery of Biomechanical Laws

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Supporting Information Text

Selection Criteria

We took an approach to choose the single ‘best’ fitting equation (f_{SR}), one that balances accuracy with parsimony, from the list of equations returned from symbolic regression (SR). First, we took the natural logarithm of the loss estimates, $\log_{10}(\text{loss})$, at each equation complexity. Then, to assess the change of $\log(\text{loss})$ across equation size, we calculated the absolute differences between consecutive $\log(\text{loss})$, $\Delta \log(\text{loss})$. The equation with the greatest $\Delta \log(\text{loss})$, meaning the greatest loss estimate improvement compared to its less complex neighboring equation, was chosen as f_{SR} for each SR trial. This approach allowed us to choose parsimonious yet accurate equations when SR was implemented in both the simulation experiment and the empirical experiment.

Equation Generalization

To evaluate how well f_{SR} can explain and predict each walking trial, f_{SR} were transformed into their generalized forms (e.g. $\ddot{\theta}_{right\ thigh} = -70.427\theta_{right\ thigh}\theta_{left\ thigh} - 36.383\theta_{right\ thigh}$ was transformed into $\ddot{\theta}_{right\ thigh} = \alpha_1\theta_{right\ thigh}\theta_{left\ thigh} + \alpha_2\theta_{right\ thigh} + \alpha_0$). During the exploration of equations, SR tends to add trivial terms that have minimal influence on the overall dynamics but marginally enhance the goodness of fit compared to the original time series. Thus, trivial terms, terms with coefficients less than 0.0001, were removed from f_{SR} . Then, we generalized our equations by replacing the constants and all coefficients with general terms (e.g. $\alpha_{t0} - \alpha_{t2}, \alpha_{s0} - \alpha_{s2}, \alpha_{f0} - \alpha_{f2}$ in Table S2).

Post-hoc Analysis on Discovered Equations of Lorenz System

We performed additional analyses to investigate how long the behavior of the original Lorenz system can be predicted based on f_{SR} . We computed the time it takes for the R^2 between the original and the predicted time series to drop below 0.995. The results showed that, on average, R^2 between the original and the predicted time series of x , y , and z maintained values above 0.995, with durations of 11.185, 10.953, 11.01 seconds, respectively (Figure 3 middle panel). If the exact coefficients were used before equation generalization, the predicted time series would have perfectly aligned with the original time series. However, the coefficients estimated from SRs are often close, but not identical, to their original counterparts, therefore reducing the prediction accuracy precipitously over time. In the discussion, we suggest and describe how making simplifying assumptions of f_{SR} can be used to overcome the issues with imperfect coefficients.

Even in their non-simplified forms, the rediscovered equations of the Lorenz system were still able to accurately predict the original trajectories for 10~11seconds (2,000~2,200 datapoints; the details on this analysis can be found in supplementary materials). This lengthy prediction, compared to other modeling techniques, is noteworthy. Several studies predicting the Lorenz system using reservoir computing trained their models with 5,000 data points and were able to successfully predict ~200-540 data points (1–3). One study that worked on optimizing reservoir computing was able to enhance the prediction up to 1240 data points (1). Other studies with temporal convolutional networks trained their models on 7,976 and 40,000 data points and were able to accurately predict 1,994 and 50,000 data points, respectively ($R^2 > 0.999$) (4, 5). In our case, when SR on the Lorenz system was performed based on a smaller number of data points

(1,000 datapoints), the prediction results outperformed those from reservoir computing and were comparable to some of the temporal convolutional network approaches.

Gait Kinematics Data Processing

We used 1946 overground walking trials suitable for symbolic regression and extracted bilateral thigh, shank, and foot segment angle time series in the sagittal plane. The extracted time series were then low pass filtered using a 4th order Butterworth filter with a cutoff frequency at 10Hz. Then, the filtered segment angle time series were differentiated to obtain segment angular velocity and acceleration time series. Based on the right heel contact information, segment angle, angular velocity, and angular acceleration time series were segmented into each gait cycle. The segmented time series were then time normalized using spline interpolation to fit within 101 datapoints. Finally, each segment angle, angular velocity, and angular acceleration time series were ensemble averaged across all walking trials and gait cycles to retrieve the typical gait kinematics of all participants in a single gait cycle.

Comparison of Original Dynamical Systems and Dynamical Systems Predicted from Symbolic Regression

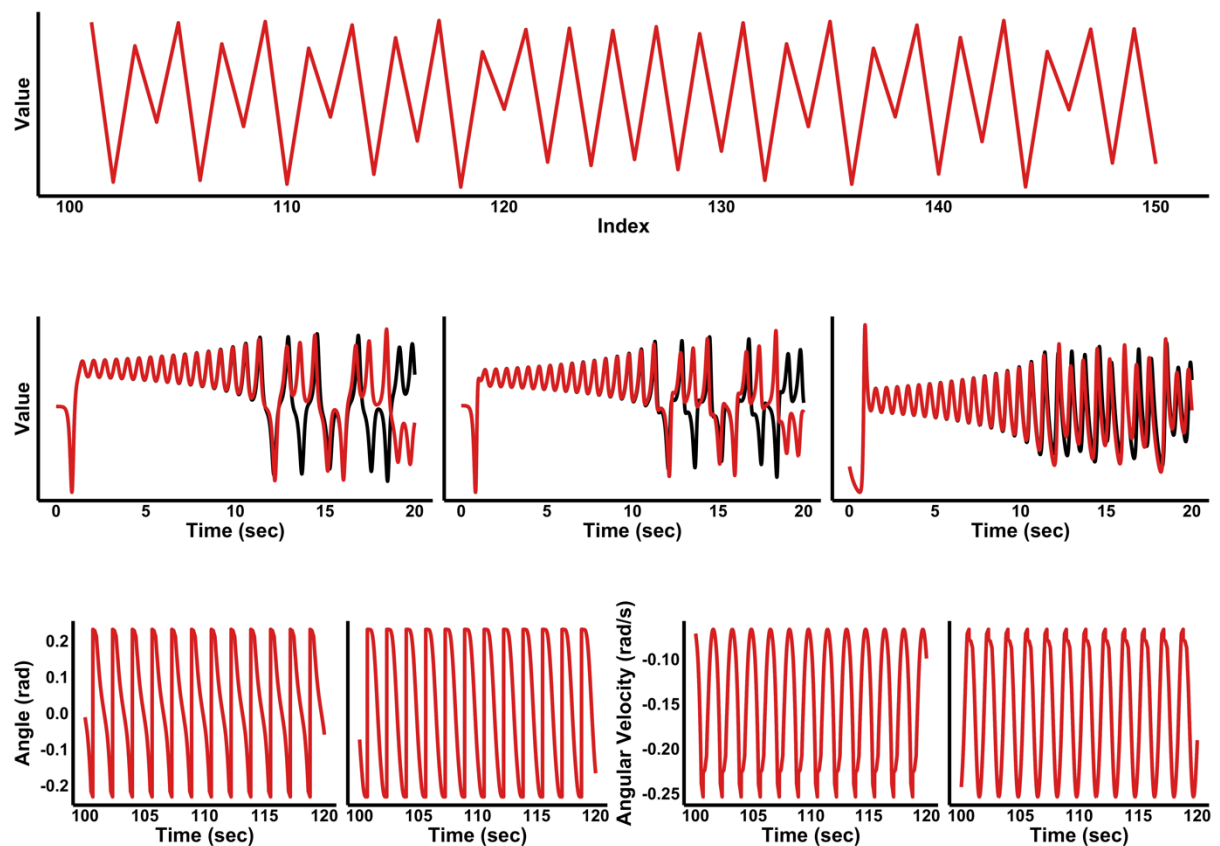


Figure S1. Original and Predicted Dynamic Systems. (Top) The original and predicted trajectories of logistic map. (Middle) The original and predicted trajectories of Lorenz system. Subfigures show results for the x, y, and z components from left to right. (Bottom) The original and predicted

kinematics of the passive dynamic walker model. Subfigures show results for θ , ϕ , $\dot{\theta}$, and $\dot{\phi}$ from left to right. Black trajectories depict the original logistic map. Red trajectories depict the predicted logistic map.

Bland Altman Analysis of Differences between Predicted and Original Hurst Exponents

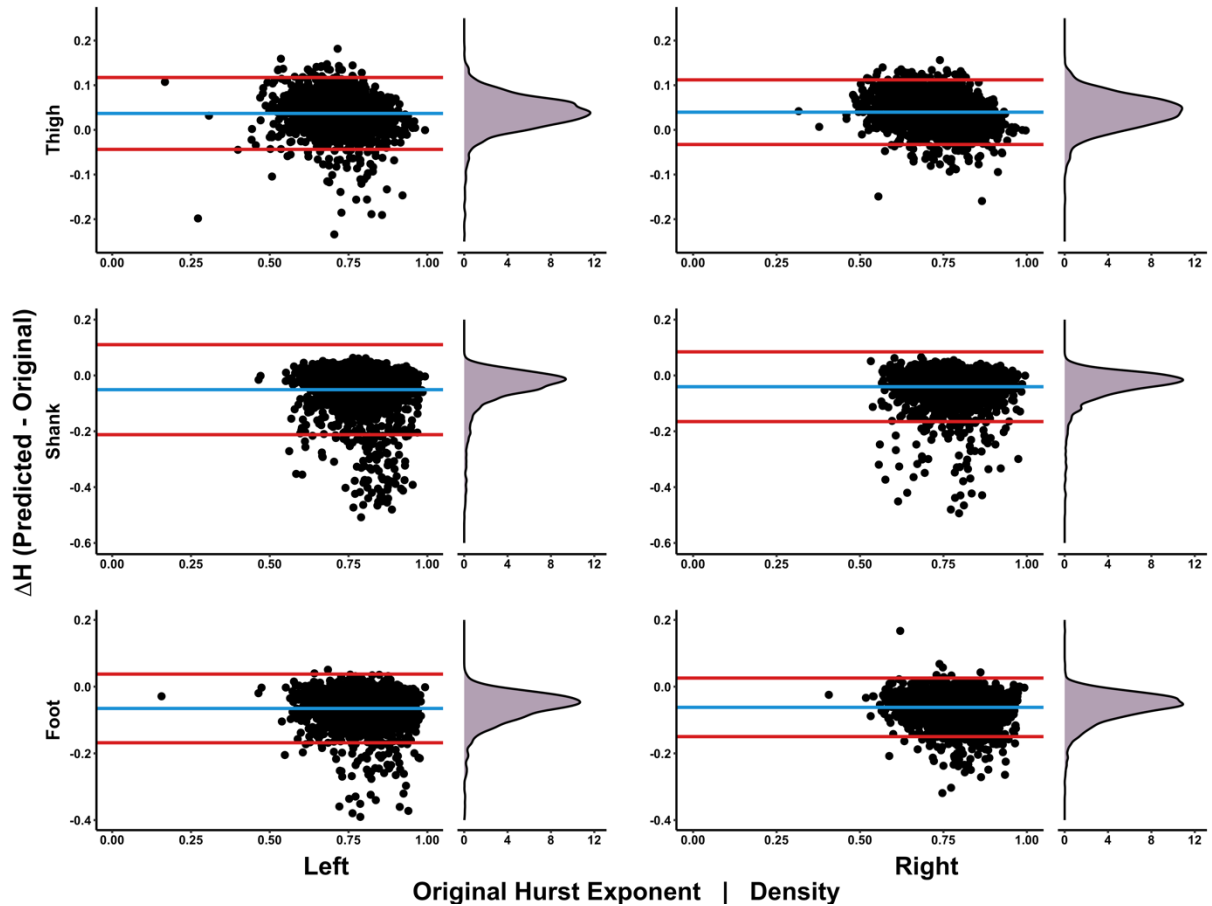


Figure S2. Bland-Altman plots and probability density functions of the differences between the predicted and the original Hurst exponents for each segment. Black dots represent the Hurst exponents of each walking trial. Blue lines indicate the mean of the differences for each segment. Red lines indicate the limits of agreement (mean $\pm$ 1.96 standard deviation). Vertically aligned probability density functions represent the distributions of the differences for each segment.

Mathematical Interpretation of Equations of Motion Discovered in Empirical Experiment

Thigh

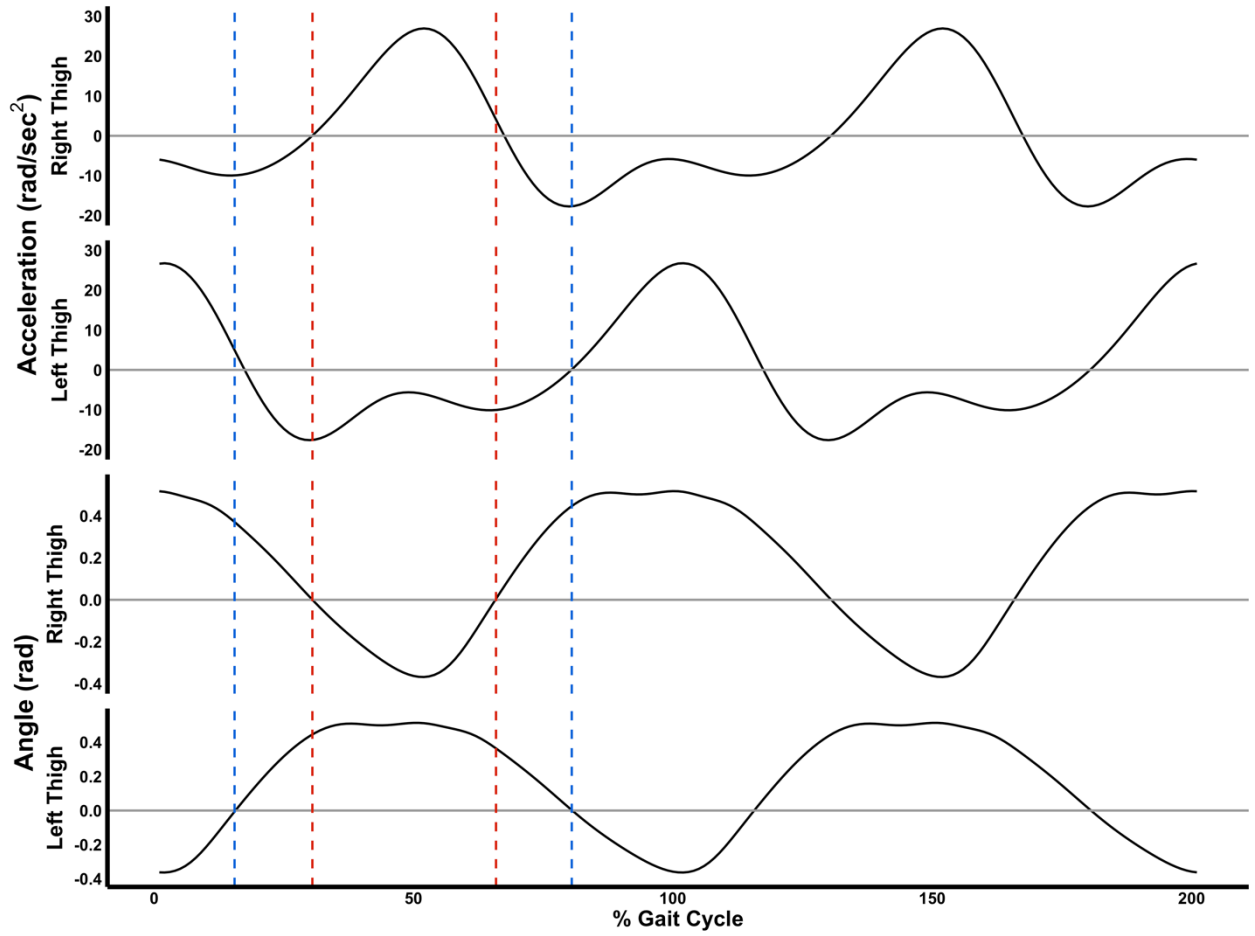


Figure S3. Typical Thigh Kinematics during Gait Cycle. Positive values indicate posterior tilt and negative values indicate anterior tilt of the thigh segment. Red dashed line indicates when $\theta_{ipsi.thigh}$ is equal to 0. Blue dashed line indicates when $\theta_{contra.thigh}$ is equal to 0. Gray solid lines indicate 0.

$$\ddot{\theta}_{thigh} = \theta_{ipsi.thigh}(\alpha_{t1}\theta_{contra.thigh} + \alpha_{t2}) + \alpha_{t0} \dots (\text{eq.1})$$

$$k = \alpha_{t1} \cdot \theta'_{contra.thigh} + \alpha_{t2} \dots (\text{eq.2})$$

To understand the physical meaning of each component in f_{SR} of $\ddot{\theta}_{thigh}$, we first focus on the intercept (α_{t0}). According to eq.1, when k equals 0, α_{t0} equates to $\ddot{\theta}_{thigh}$. If $\theta_{ipsi.thigh}$ equals 0 or $\theta_{contra.thigh}$ equals $-\alpha_{t2}/\alpha_{t1}$ in eq.2, k equals 0. Since the domain of $\theta_{ipsi.thigh}$ and $\theta_{contra.thigh}$ is approximately $[-0.36, 0.51]$, there exists $\theta_{ipsi.thigh}$ such that $\theta_{ipsi.thigh}$ is equal to 0. Based on the mean values of α_{t1} and α_{t2} found from SR, the following assumption can be made.

$$-\alpha_{t2}/\alpha_{t1} \approx (-36.383)/(-70.437) \approx -0.5165 \dots (\text{eq.3})$$

Due to the domain of $\theta_{contra.thigh}$, there does not exist $\theta_{contra.thigh}$ such that $\theta_{contra.thigh}$ is equal to $-\alpha_{t2}/\alpha_{t1}$. Therefore, α_{t0} equals $\ddot{\theta}_{thigh}$ if and only if $\theta_{ipsi.thigh}$ is 0 (Figure S2 red dashed lines at 30 – 41% and 65 – 66% of gait cycle). That is, α_{t0} reflects $\ddot{\theta}_{thigh}$ when the ipsilateral thigh is perpendicular to the surface.

Then, to aid the physical interpretation of coefficients included in f_{SR} of $\ddot{\theta}_{thigh}$, we let $\theta_{contra.thigh}'$ be an arbitrary constant that represents $\theta_{contra.thigh}$ at an arbitrary time during the gait cycle. We substitute $\theta_{contra.thigh}$ with $\theta_{contra.thigh}'$ and insert eq.4 into eq.1 to get eq.5.

$$k' = \alpha_{t1} \cdot \theta_{contra.thigh}' + \alpha_{t2} \dots (\text{eq.4})$$

$$\ddot{\theta}_{thigh} = k' \cdot \theta_{ipsi.thigh} + \alpha_{t0} \dots (\text{eq.5})$$

Based on eq.5, we note that k' reflects the linear relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{thigh}$ at an arbitrary time. Specifically, when $\theta_{contra.thigh}'$ equals 0, k equates to α_{t2} , meaning that α_{t2} reflects the linear relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{thigh}$ at the moment $\theta_{contra.thigh}'$ becomes 0 (Figure S2 blue dashed lines at 15 – 16% and 80 – 81% of gait cycle). To rephrase, α_{t2} portrays the linear relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{thigh}$ when the contralateral thigh is perpendicular to the surface. Now, if we return the constant $\theta_{contra.thigh}'$ to the time dependent variable $\theta_{contra.thigh}$, k' also becomes a time dependent variable, k , that follows the temporal variation of $\theta_{contra.thigh}$. Hence, we can state that the relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{thigh}$, symbolized as k , is amplified and diminished according to the change of $\theta_{contra.thigh}$ over time. Furthermore, such temporal trends of the relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{thigh}$ are scaled by α_{t1} , inferring that α_{t1} scales the impact of the contralateral thigh on the relationship between $\theta_{ipsi.thigh}$ and $\ddot{\theta}_{thigh}$.

Shank

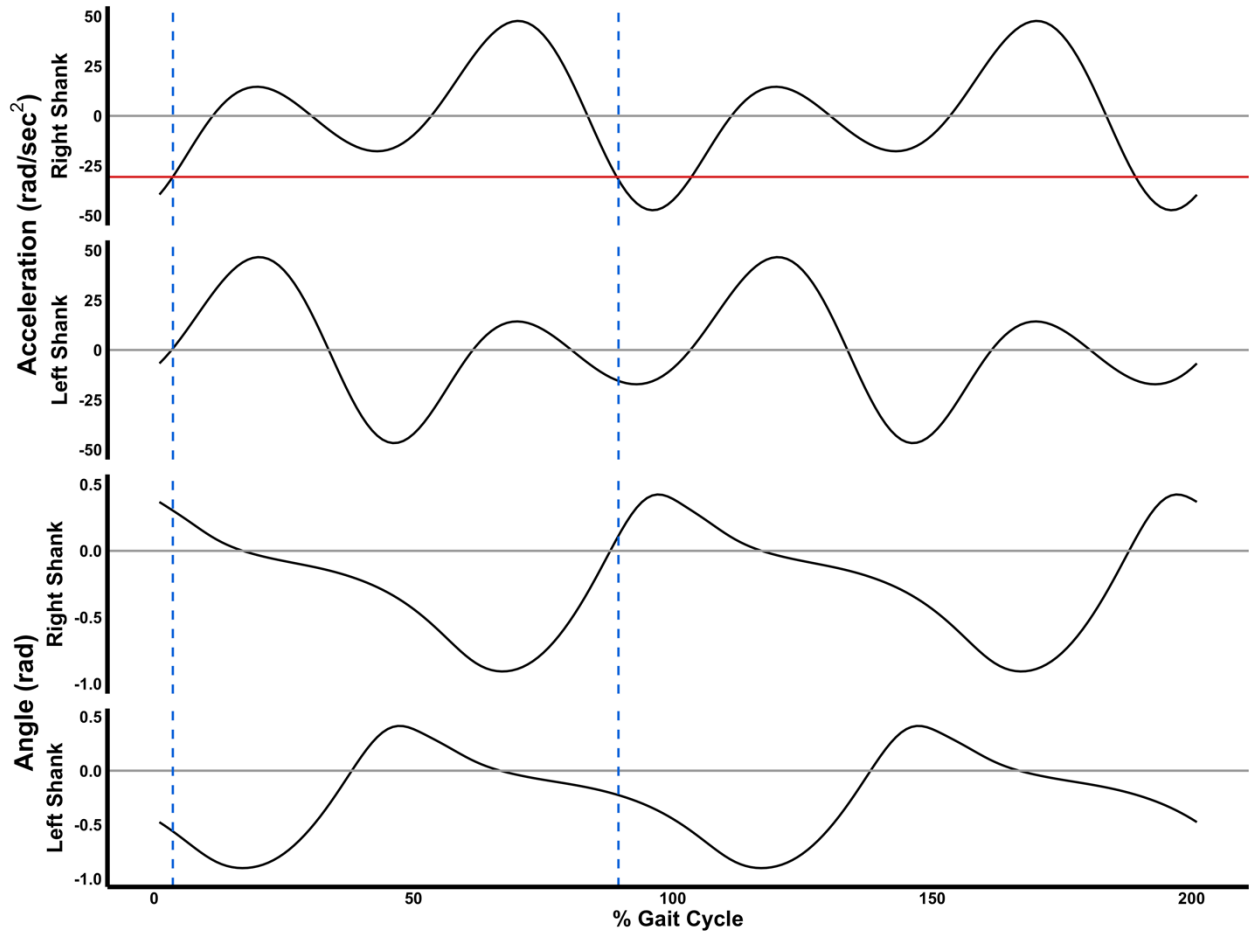


Figure S4. Typical Shank Kinematics during Gait Cycle. Positive values indicate posterior tilt and negative values indicate anterior tilt of the shank segment. Red horizontal line indicates α_{s0} when α_{s0} equates to $\ddot{\theta}_{shank}$. Blue dashed line indicates when $\theta_{ipsi.shank}/\theta_{contra.shank}$ equals $-\alpha_{s2}/\alpha_{s1}$. Gray solid lines indicate 0.

$$\ddot{\theta}_{shank} = \alpha_{s1}\theta_{ipsi.shank} + \alpha_{s2}\theta_{contra.shank} + \alpha_{s0} \dots (\text{eq.6})$$

$$l = \alpha_{s1}\theta_{ipsi.shank} + \alpha_{s2}\theta_{contra.shank} \dots (\text{eq.7})$$

Again, to elucidate the physical meanings of the components in f_{SR} of $\ddot{\theta}_{shank}$, we start with explaining the intercept (α_{s0}). According to eq.6 and eq.7, if l equals 0, α_{s0} equates to $\ddot{\theta}_{shank}$. If $\theta_{ipsi.shank}$ is 0 and $\theta_{contra.shank}$ is 0, or if $\theta_{ipsi.shank}/\theta_{contra.shank}$ equals $-\alpha_{s2}/\alpha_{s1}$, l equals 0. However, during a typical gait cycle, there does not exist any moment when $\theta_{ipsi.shank}$ and $\theta_{contra.shank}$ both equal 0. However, there exists two moments in a typical gait cycle when $\theta_{ipsi.shank}/\theta_{contra.shank}$ become $-\alpha_{s2}/\alpha_{s1}$ ($\theta_{ipsi.shank} \approx 0.3057, \theta_{contra.shank} \approx -0.5576$; $\theta_{ipsi.shank} \approx 0.0781, \theta_{contra.shank} \approx -0.2188$, Figure S3 blue dashed lines at 3 – 4% and 89 – 90% of gait cycle, respectively). Hence, if and only if $\theta_{ipsi.shank}/\theta_{contra.shank}$ becomes $-\alpha_{s2}/\alpha_{s1}$, m becomes 0 and α_{s0} is equal to $\ddot{\theta}_{shank}$ (Figure S3 intersection between red horizontal line and blue

dashed lines in top panel). To determine which physical situations $\theta_{ipsi.shank}/\theta_{contra.shank}$ become $-\alpha_{s2}/\alpha_{s1}$, the physical meanings of both coefficients should be clarified.

Unlike f_{SR} of $\ddot{\theta}_{thigh}$, there are no products between $\theta_{ipsi.shank}$ and $\theta_{contra.shank}$ in f_{SR} of $\ddot{\theta}_{shank}$. This indicates that $\theta_{ipsi.shank}$ and $\theta_{contra.shank}$ independently influence $\ddot{\theta}_{shank}$. Thus, we can state that α_{s1} and α_{s2} reflect how $\theta_{ipsi.shank}$ and $\theta_{contra.shank}$ linearly relate to $\ddot{\theta}_{shank}$, respectively. Moreover, $\alpha_{s1}\theta_{ipsi.shank}$ and $\alpha_{s2}\theta_{contra.shank}$ each refer to the weighted impact of each segment angle on $\ddot{\theta}_{shank}$. This suggests $\theta_{ipsi.shank}/\theta_{contra.shank}$ is equal to $-\alpha_{s2}/\alpha_{s1}$, which is mathematically equivalent to l equating to 0 when the weighted impacts of bilateral shank angles add to 0. Therefore, α_{s0} reflects $\ddot{\theta}_{shank}$ when the net impact of bilateral shank angles on $\ddot{\theta}_{shank}$ is nullified.

Foot

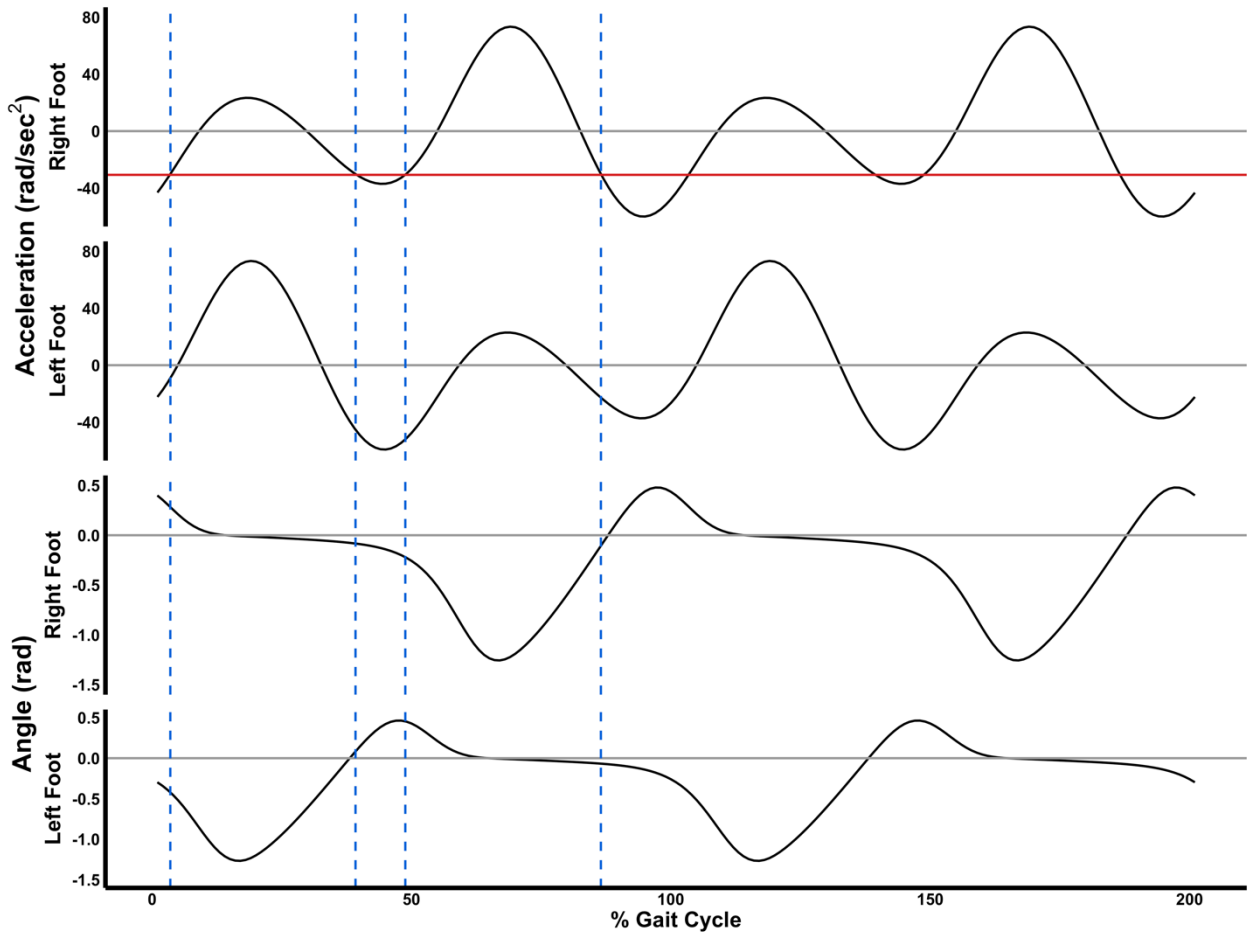


Figure S5. Typical Foot Kinematics during Gait Cycle. Positive values indicate posterior tilt and negative values indicate anterior tilt of the foot segment. Red horizontal line indicates α_{f0} when α_{f0} equates to $\ddot{\theta}_{foot}$. Blue dashed line indicates when $\theta_{ipsi.foot}/\theta_{contra.foot}$ equals $-\alpha_{f2}/\alpha_{f1}$. Gray solid lines indicate 0.

974

975 $\ddot{\theta}_{foot} = \alpha_{f1}\theta_{ipsi.foot} + \alpha_{f2}\theta_{contra.foot} + \alpha_{f0} \dots (\text{eq.8})$

976 $m = \alpha_{f1}\theta_{ipsi.foot} + \alpha_{f2}\theta_{contra.foot} \dots (\text{eq.9})$

977

978 $\ddot{\theta}_{shank}$ and $\ddot{\theta}_{foot}$ trajectories resemble each other and f_{SR} of $\ddot{\theta}_{shank}$ and $\ddot{\theta}_{foot}$ have the same
979 structure. Therefore, the physical, mathematical interpretation, f_{SR} of $\ddot{\theta}_{foot}$, is akin to that of
980 $\ddot{\theta}_{shank}$. Based on eq.8 and eq.9, if m equals 0, α_{f0} becomes $\ddot{\theta}_{foot}$. If $\theta_{ipsi.foot}$ and $\theta_{contra.foot}$ are
981 both equal to 0, or if $\theta_{ipsi.foot}/\theta_{contra.foot}$ is $-\alpha_{f2}/\alpha_{f1}$, m equates to 0. There does not exist any
982 moment at which both $\theta_{ipsi.foot}$ and $\theta_{contra.foot}$ are 0 at the same time, whereas there are four
983 moments when $\theta_{ipsi.foot}/\theta_{contra.foot}$ equals $-\alpha_{f2}/\alpha_{f1}$ ($\theta_{ipsi.foot} \approx 0.2826, \theta_{contra.foot} \approx -0.4224$;
984 $\theta_{ipsi.foot} \approx 0.1043, \theta_{contra.foot} \approx -0.0855$; $\theta_{ipsi.foot} \approx -0.2127, \theta_{contra.foot} \approx 0.4575$; $\theta_{ipsi.foot} \approx$
985 $-0.1060, \theta_{contra.foot} \approx -0.0675$, Figure S4 blue dashed lines at 3 – 4%, 39 – 40%, 48 – 49%,
986 and 86 – 76% of gait cycle, respectively). Thus, if and only if $\theta_{ipsi.foot}/\theta_{contra.foot}$ becomes
987 $-\alpha_{f2}/\alpha_{f1}$, α_{f0} is equal to $\ddot{\theta}_{foot}$.

988

989 Just like f_{SR} of $\ddot{\theta}_{shank}$, $\theta_{ipsi.foot}$ and $\theta_{contra.foot}$ independently influence $\ddot{\theta}_{foot}$, meaning α_{f1} and α_{f2}
990 reflect the linear relationship of each foot angle and $\ddot{\theta}_{foot}$. Furthermore, $\alpha_{f1}\theta_{ipsi.foot}$ and
991 $\alpha_{f2}\theta_{contra.foot}$ each refer to the weighted impact of each foot angle on $\ddot{\theta}_{foot}$. This indicates that
992 $\theta_{ipsi.foot}/\theta_{contra.foot}$ equals $-\alpha_{f2}/\alpha_{f1}$, or m equals 0, when the weighted influences of bilateral
993 shank angles sum to 0. Consequently, α_{f0} represents $\ddot{\theta}_{foot}$ when the net impact of bilateral foot
994 angles on $\ddot{\theta}_{foot}$ are canceled out.

Table S1. Equations Discovered by Symbolic Regression on Each Dynamical System and Passive Dynamic Walker Model

Systems	Original Equations	Discovered Equations (f_{SR})	Number of Instances
Logistic Map	$x_{t+1} = 3.6x_t(1 - x_t)$	$x_{t+1} = -3.6x_t^2 + 3.6x_t$	17
		$x_{t+1} = 3.6x_t(1 - x_t)$	3
Lorenz System (dx/dt)	$dx/dt = 10(y - x)$	$dx/dt = 9.996(y - x)$	20
Lorenz System (dy/dt)	$dy/dt = x(28 - z) - y$	$dy/dt = x(28.000 - z) - y$	20
Lorenz System (dz/dt)	$dz/dt = xy - \frac{8}{3}z$	$dz/dt = xy - 2.666z$	18
		$dz/dt = xy - 2.670z + 0.102$	1
		$dz/dt = xy - 2.672z + 0.159$	1
Passive Dynamic Walker ($\ddot{\theta}$)	$\ddot{\theta} = \sin(\theta - 0.014)$	$\ddot{\theta} = \sin(\theta - 0.014)$	20
Passive Dynamic Walker ($\ddot{\phi}$)	$\ddot{\phi} = \ddot{\theta} + \dot{\theta}^2 \sin(\phi) - \cos(\theta - 0.014) \sin(\phi)$	$\ddot{\phi} = -2.519\dot{\phi}(\dot{\theta} + 0.397)$	17
		$\ddot{\phi} = (\dot{\theta}^2 - \cos(\theta - 0.014)) \sin(\phi) + \sin(\theta - 0.014)$	2
		$\ddot{\phi} = 1.002\dot{\theta} \frac{\phi}{-0.173 + \frac{0.701}{\theta}} + 1.002\theta - 1.002\sin(\dot{\theta}) - 0.014$	1

Table S2. Generalized Forms of Equations Discovered by Symbolic Regression on Each Segment

Systems	Discovered Equations (f_{SR})	Equation Complexity	Number of Instances
Thigh	$\ddot{\theta}_{thigh} = \theta_{ipsi.thigh}(\alpha_{t1}\theta_{contra.thigh} + \alpha_{t2}) + \alpha_{t0}$	9	20
Shank	$\ddot{\theta}_{shank} = \alpha_{s1}\theta_{ipsi.shank} + \alpha_{s2}\theta_{contra.shank} + \alpha_{s0}$	9	20
Foot	$\ddot{\theta}_{foot} = \alpha_{f1}\theta_{ipsi.foot} + \alpha_{f2}\theta_{contra.foot} + \alpha_{f0}$	9	19
	$\ddot{\theta}_{foot} = \alpha_{f1}\theta_{ipsi.foot} + \alpha_{f2}\theta_{contra.foot} + \alpha_{f3}\theta_{ipsi.shank} + \alpha_{f0}$	13	1
α_{ti} : ith coefficient included in thigh f_{SR}			
α_{si} : ith coefficient included in shank f_{SR}			
α_{fi} : ith coefficient included in foot f_{SR}			
α_{t0} : Intercept included in thigh f_{SR}			
α_{s0} : Intercepts included in shank f_{SR}			
α_{f0} : Intercepts included in foot f_{SR}			
$\ddot{\theta}_{thigh}$: Thigh angular acceleration / $\ddot{\theta}_{shank}$: Shank angular acceleration / $\ddot{\theta}_{foot}$: Foot angular acceleration			
$\theta_{ipsi.thigh}$: Ipsilateral thigh angle / $\theta_{contra.thigh}$: Contralateral thigh angle			
$\theta_{ipsi.shank}$: Ipsilateral shank angle / $\theta_{contra.shank}$: Contralateral shank angle			
$\theta_{ipsi.foot}$: Ipsilateral foot angle / $\theta_{contra.foot}$: Contralateral foot angle			

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Table S3. Initial Conditions and Parameters Used in Simulations.

Systems	Equations	Initial Conditions	Control Parameters	Data Points (Sampling Rate)
Logistic Map	$x_{t+1} = \alpha x_t(1 - x_t)$	$x_0 = 0.6$	$\alpha = 3.6$	1000 (No sampling rate)
Lorenz System	$\frac{dx}{dt} = \rho(y - x)$	$x_0 = 0.6$	$\sigma = 10$	1000 (0.005)
	$\frac{dy}{dt} = x(\sigma - z) - y$	$y_0 = -0.01$	$\rho = 28$	
	$\frac{dz}{dt} = xy - \beta z$	$z_0 = 9$	$\beta = 8/3$	
Passive Dynamic Walker	$\ddot{\theta} = \sin(\theta - \gamma)$	$\theta_0 = 0.231$	$\gamma = 0.014$	330 (0.005)
	$\ddot{\phi} = \ddot{\theta} + \dot{\theta}^2 \sin(\phi) - \cos(\theta - \gamma) \sin(\phi)$	$\dot{\theta}_0 = -0.227$		
		$\phi_0 = 0.463$		
		$\dot{\phi}_0 = -0.024$		

998

999

Table S4. Input Variables and Outcome Variables for Each Dynamical System and Passive Dynamic Walker Model.

Systems	Dependent Variable	Independent Variable
Logistic Map	x_{t+1}	x_t
Lorenz System	dx/dt	x, y, z
	dy/dt	x, y, z
	dz/dt	x, y, z
Passive Dynamic Walking Model	$\ddot{\theta}$	$\theta, \dot{\theta}, \phi, \dot{\phi}$
	$\ddot{\phi}$	$\theta, \dot{\theta}, \phi, \dot{\phi}$

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Table S5. Parameters Used for Symbolic Regression on Dynamical Systems and Passive Dynamic Walker Model.

Parameters of Symbolic Regression	Values Used
Number of Individuals per Population	33 individuals/population
Number of Populations	15 populations
Number of Cycles per Iteration	550 cycles/iteration
Number of Iterations	100k iterations (Logistic map, Lorenz system) 300k iterations (Passive dynamic walking model)
Max Equation Complexity	20 (Logistic map, Lorenz system, Passive dynamic walking model – $\ddot{\theta}$) 50 (Passive dynamic walking model – $\ddot{\phi}$)

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Table S6. Subject Characteristics

Age Group (Age)	N Subjects	N Trials	# Female	Age (years)	Height (cm)	Weight (kg)
Young (19-35)	35	606	17	24.60 $\pm$ 2.70	173.73 $\pm$ 7.76	72.44 $\pm$ 15.04
Middle (36-55)	36	635	29	46.69 $\pm$ 5.84	171.52 $\pm$ 7.99	84.18 $\pm$ 18.07
Old (56+)	41	705	21	64.73 $\pm$ 7.51	170.89 $\pm$ 9.85	81.08 $\pm$ 17.76

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1005

Table S7. Parameters Used for Symbolic Regression on Gait Data

Parameters of Symbolic Regression	Values Used
Number of Individuals per Population	33 individuals/population
Number of Populations	15 populations
Number of Cycles per Iteration	550 cycles/iteration
Number of Iterations	300k iterations
Max Equation Complexity	50

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1007
1008

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